01 / LEARNING GUIDE

# Engineering & Math Theory

EX28 · Numerical algebra · Matrices, dimensions and products

> **Reading order:** 01 Engineering & Math Theory → 02 Python Reference → 03 Exercise → 04 Pseudocode → programming. This guide belongs to EX28, class 18.

## 1. What is a matrix?

A **matrix** is a rectangular arrangement of numbers organized into rows and columns. It represents coefficients, transforms vectors, and describes linear relationships. We begin with mathematical rules before assigning physical meaning.

$$A = \begin{pmatrix} a_{11} & a_{12} & \cdots & a_{1n} \\ a_{21} & a_{22} & \cdots & a_{2n} \\ \vdots & \vdots & \ddots & \vdots \\ a_{m1} & a_{m2} & \cdots & a_{mn} \end{pmatrix} \in \mathbb{R}^{m \times n}$$

**m × n** means m rows and n columns. The entry $a_{ij}$ lies in row i, column j. A matrix may be rectangular.

$$C = \begin{pmatrix} 2 & -1 & 3 \\ 0 & 4 & 5 \end{pmatrix} \in \mathbb{R}^{2 \times 3}$$

Matrix C has 2 rows, 3 columns and 6 entries. The row-2, column-3 entry is 5.

## 2. Vectors, matrices and dimensionality

A mathematical vector can be displayed as a column. In NumPy, we initially use a **one-dimensional array** with shape **(n,)**, rather than a two-dimensional column array of shape **(n,1)**. We will distinguish these representations later.

$$\boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}, \qquad A \in \mathbb{R}^{2 \times 2}$$

We use **bold letters** for vectors, uppercase letters for matrices, and ordinary letters or subscripts for scalar entries.

## 3. Matrix addition and scalar multiplication

Two matrices may be added only if their shapes are identical. Both addition and scalar multiplication operate entry by entry.

$$(A + B)_{ij} = a_{ij} + b_{ij}, \qquad (\alpha A)_{ij} = \alpha a_{ij}$$

Manual example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad B = \begin{pmatrix} 2 & 0 \\ -1 & 3 \end{pmatrix}$$

$$A + B = \begin{pmatrix} 3 & 2 \\ 2 & 7 \end{pmatrix}, \qquad 2A = \begin{pmatrix} 2 & 4 \\ 6 & 8 \end{pmatrix}$$

## 4. Transpose

The transpose exchanges rows and columns. If A is m × n, then $A^T$ is n × m.

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix} \quad \Longrightarrow \quad A^T = \begin{pmatrix} 1 & 3 \\ 2 & 4 \end{pmatrix}$$

Useful properties include $(A^T)^T = A$ and $(A+B)^T = A^T + B^T$ when addition is defined.

## 5. Matrix multiplication: the key rule

Matrix multiplication is **not** the element-wise multiplication of matching positions. Entry (i,j) of AB is the dot product of row i of A with column j of B.

$$(AB)_{ij} = \sum_{k=1}^{n} a_{ik} b_{kj}$$

If A has shape m × n and B has shape n × p, then AB is defined and has shape m × p.

$$(m \times n)\,(n \times p) \longrightarrow (m \times p)$$

Manual calculation using the matrices above:

$$AB = \begin{pmatrix} 1(2) + 2(-1) & 1(0) + 2(3) \\ 3(2) + 4(-1) & 3(0) + 4(3) \end{pmatrix}$$

$$AB = \begin{pmatrix} 0 & 6 \\ 2 & 12 \end{pmatrix}$$

### Why is AB generally different from BA?

Reversing the factors changes which rows and columns are combined. Even when both products exist, their results usually differ.

$$BA = \begin{pmatrix} 2 & 4 \\ 8 & 10 \end{pmatrix} \neq AB$$

Matrix multiplication is **associative** for compatible sizes, but generally **not commutative**.

## 6. Matrix-vector multiplication

Multiplying A by a vector **x** forms a linear combination of the columns of A. The components of **x** supply the weights.

$$A = \begin{pmatrix} a_{11} & a_{12} \\ a_{21} & a_{22} \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} x_1 \\ x_2 \end{pmatrix}$$

$$A\boldsymbol{x} = x_1 \begin{pmatrix} a_{11} \\ a_{21} \end{pmatrix} + x_2 \begin{pmatrix} a_{12} \\ a_{22} \end{pmatrix}$$

Fully worked algebra example:

$$A = \begin{pmatrix} 1 & 2 \\ 3 & 4 \end{pmatrix}, \quad \boldsymbol{x} = \begin{pmatrix} 2 \\ -1 \end{pmatrix}$$

$$A\boldsymbol{x} = \begin{pmatrix} 1(2) + 2(-1) \\ 3(2) + 4(-1) \end{pmatrix} = \begin{pmatrix} 0 \\ 2 \end{pmatrix}$$

## 7. Linear transformations

A matrix defines a linear map from an input vector to an output vector. A compatible linear map satisfies:

$$A(\boldsymbol{u}+\boldsymbol{v}) = A\boldsymbol{u} + A\boldsymbol{v}$$

$$A(\alpha\boldsymbol{u}) = \alpha A\boldsymbol{u}$$

A linear map can alter lengths and directions while preserving addition and scalar multiplication. Later we will use this idea for rotations, reference frames, sensors and simulations.

## 8. Physical application: two known force directions

Consider two actuators applying forces of magnitudes 10 N and 5 N along the following unit directions:

$$\boldsymbol{e}_1 = \begin{pmatrix} 0.6 \\ 0.8 \end{pmatrix}, \quad \boldsymbol{e}_2 = \begin{pmatrix} -0.8 \\ 0.6 \end{pmatrix}$$

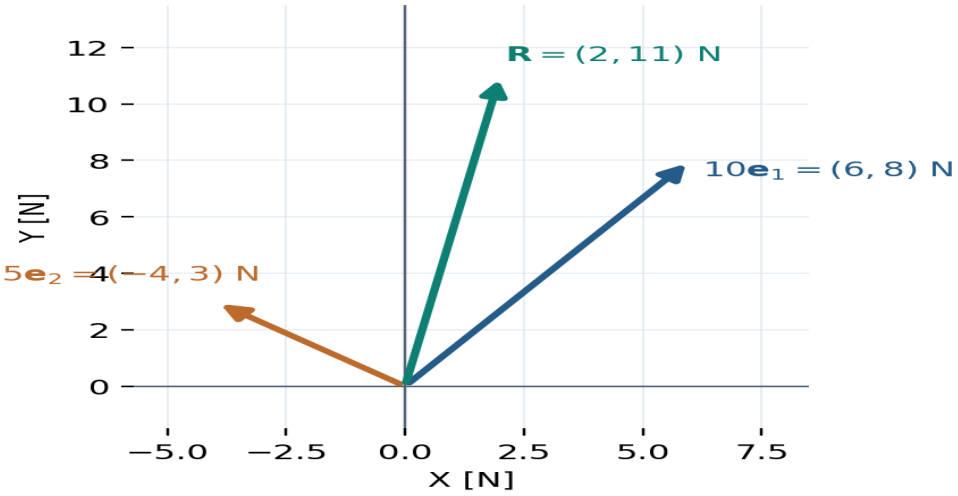
Each column of A contains a direction vector. The magnitude vector **u** contains the two force magnitudes.

$$A = \begin{pmatrix} 0.6 & -0.8 \\ 0.8 & 0.6 \end{pmatrix}, \quad \boldsymbol{u} = \begin{pmatrix} 10 \\ 5 \end{pmatrix} \text{N}$$

$$\boldsymbol{R} = A\boldsymbol{u} = 10\boldsymbol{e}_1 + 5\boldsymbol{e}_2$$

$$\boldsymbol{R} = \begin{pmatrix} 6-4 \\ 8+3 \end{pmatrix} = \begin{pmatrix} 2 \\ 11 \end{pmatrix} \text{N}$$

The resultant has magnitude $\sqrt{(2^2 + 11^2)} = \sqrt{125} \approx 11.1803$ N. Matrix A is dimensionless because its entries are direction cosines. Force units come from **u**.



## 9. Choosing the appropriate operation

| Goal | Operation |
|---|---|
| Combine equal-shape matrices | A + B |
| Exchange rows and columns | A.T |
| Scale every entry | k * A |
| Multiply entries position by position | A * B (NumPy) |
| Apply a linear transformation | A @ x, A @ B |

## 10. Identity and zero matrices

The **identity matrix** I leaves a compatible vector unchanged; the **zero matrix** O contains only zeros. These are the neutral elements of matrix multiplication and matrix addition, respectively.

$$I_2 = \begin{pmatrix} 1 & 0 \\ 0 & 1 \end{pmatrix}, \quad I_2\boldsymbol{x} = \boldsymbol{x}, \quad A + O = A$$

## 11. Rectangular matrix products

Matrices need not be square. For example, a 3 × 2 matrix maps a two-component input to a three-component output. The number of input components must match the number of columns.

$$P = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ 2 & -1 \end{pmatrix}, \quad \boldsymbol{q} = \begin{pmatrix} 3 \\ 2 \end{pmatrix}$$

$$P\boldsymbol{q} = \begin{pmatrix} 3 \\ 2 \\ 4 \end{pmatrix} \in \mathbb{R}^3$$

## 12. Scope and next steps

EX28 solves the **forward problem**: given A and **u**, calculate **R** = A**u**. We do **not** yet solve for **u** from **R**. That requires linear systems and, later, `np.linalg.solve`.

> **Key takeaway:** matrix dimensions determine whether products are valid; matrix columns describe how the inputs contribute to the output.